Version A

Question 1:

1. What are the limitations of bidirectional RNN,where you do not use it?
2. What is the purpose of the reset gate in GRU?
3. Which layer type is responsible for handling variable-length outputs in an RNN?
4. Which layer type is commonly used in RNNs for machine translation tasks?
5. What is the purpose of the teacher forcing technique in RNN training?
6. In the Transformer architecture, which is not a sequential model and uses only feed-forward networks, will the output for the input sequences
   A,B,C,D,Y
   Y,A,B,C,D
   Will "Y" output be the same or not ? Why?

Question 2:

Ewe = [0.2,0.5,0.3,0.7]
Eare = [0.4,0.6,0.4,0.5]
Egood =[0.5,0.4,0.2,0.6]
Efriend=[0.7,0.5,0.3,0.5]

W(q) = [[1 0 1][1 0 1][1 0 1]]
W(k) = [[0 1 0][0 1 0][0 1 0]]
W(v) = [[1 1 0][1 1 0][1 1 0]]

1. What are q,k,v
2. What is the self attention of Egood?